Medical Microbiology

Lecture Five

Organs and Cells of Immune System

Third stage- College of Dentistry

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Organs and Cells of the Immune System:

Lymphoid organs are specialized structures involved in immune responses.
These organs contain lymphoid cells and are classified into two main types:

1-Primary lymphoid organs.

2-Secondary lymphoid organs.

1. Primary Lymphoid Organs:

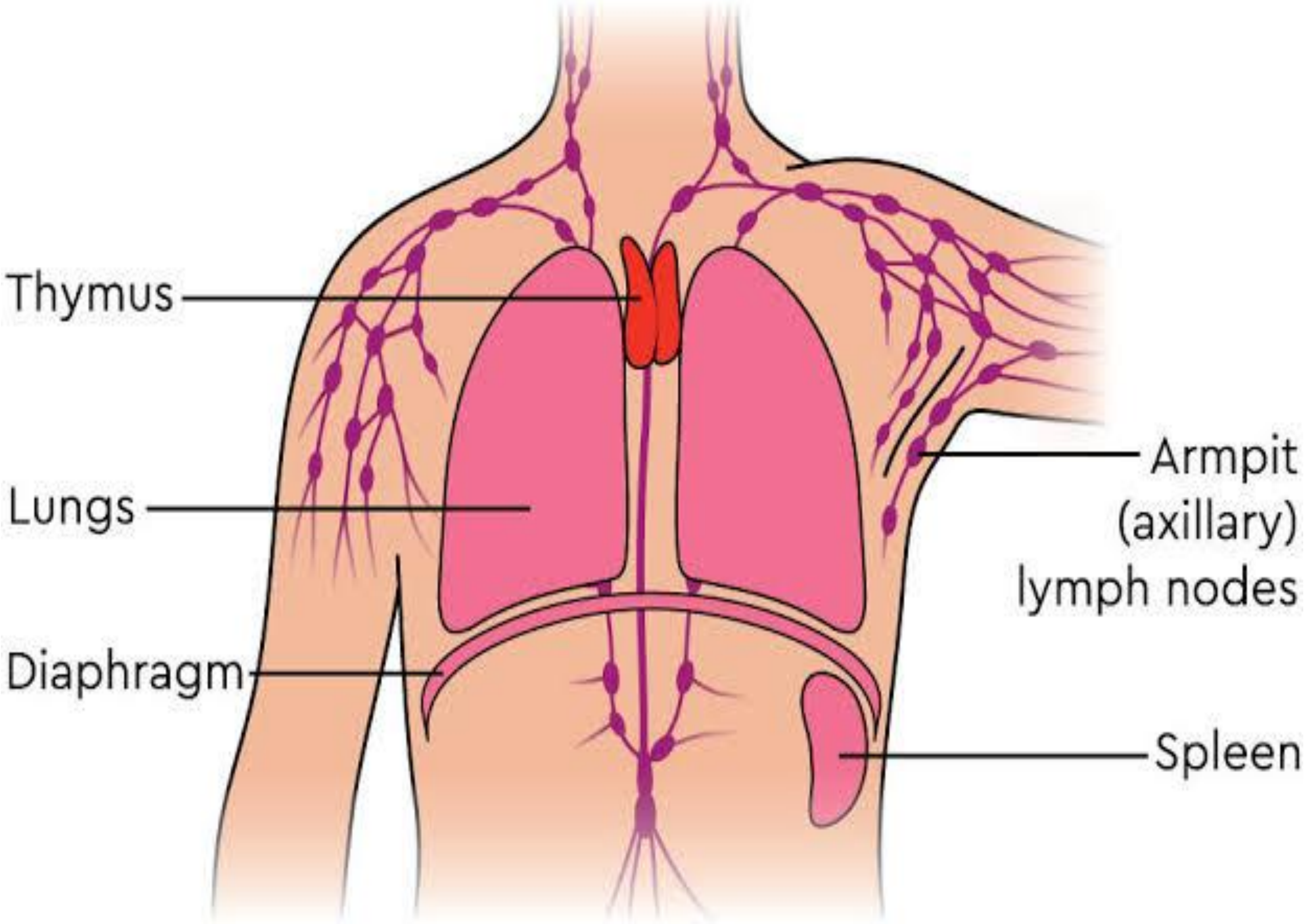
- Primary lymphoid organs are the main sites of **lymphopoiesis (**the production and development of lymphoid cells**)**.
- In these organs, lymphoid cells **proliferate, differentiate, and mature into immunocompetent cells** (capable of mounting an immune response), even **without exposure to antigens**.
- These organs are **well-developed at birth** but **shrink (atrophy) with age**.
- The major primary lymphoid organs in humans are:

A-Thymus: the site of **T cell maturation.**

B-Bone marrow: the site of **B cell maturation.**

A-Thymus:

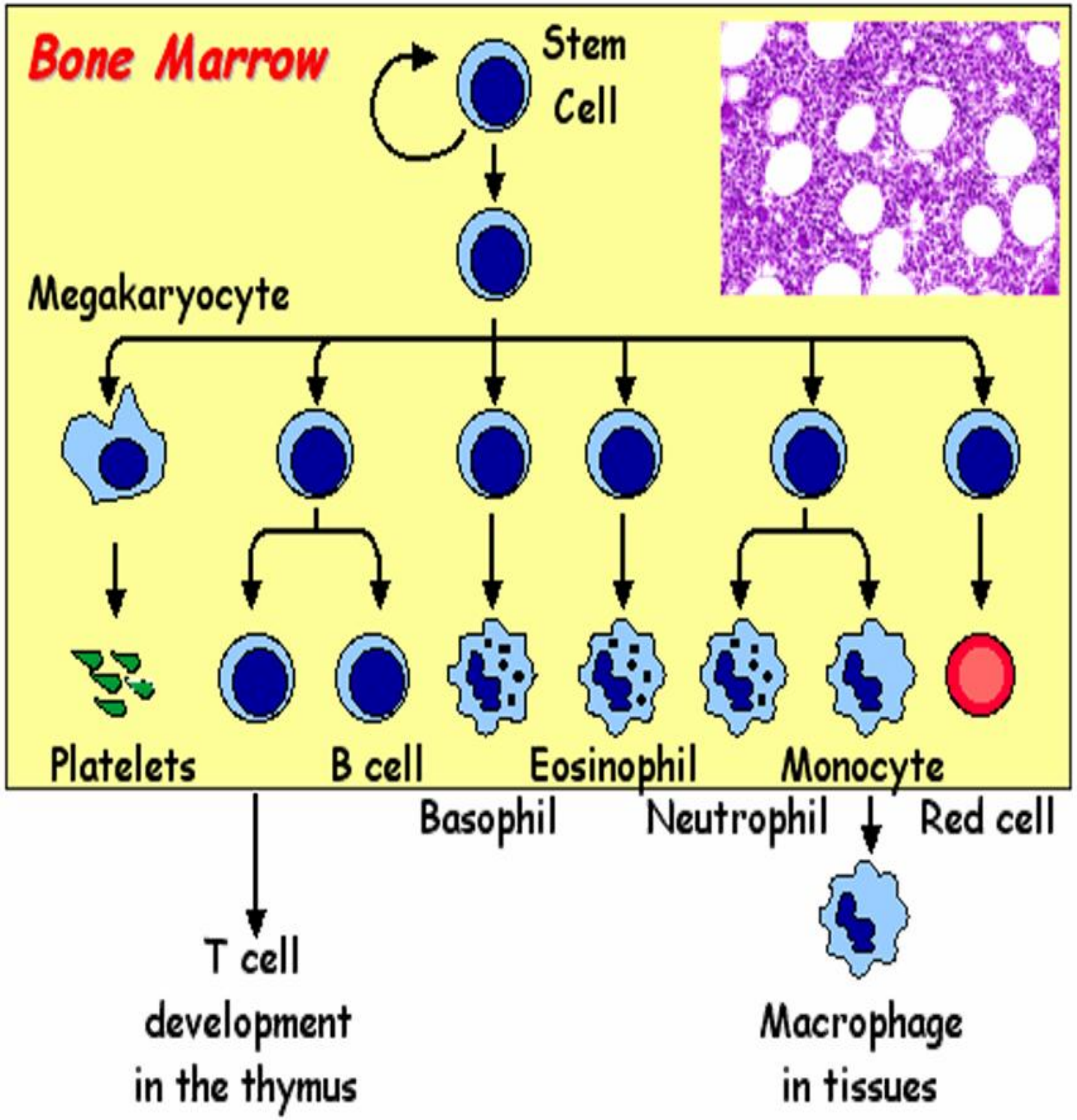
- The Thymus is the primary site for T cell differentiation and maturation. It is anatomically divided into two regions: the cortex and the medulla. Several types of cells are found within the thymus, including **stromal cells**, **epithelial cells**, **macrophages**, **dendritic cells**, and **thymocytes**.
- T cell precursors migrate from the bone marrow to the thymus, where they are referred to as **thymocytes**. These thymocytes initially cluster in the **cortex**, where they begin their development. As they mature, they migrate toward the **medulla**, undergoing further differentiation.
- During this process, thymocytes acquire specific surface molecules known as **clusters of differentiation (CD)**, which help define their function and stage of development, for example, **CD3, CD4 and CD8**.


Thymus
Lungs
Diaphragm
Armpit
(axillary)
lymph nodes
Spleen

- In **cortex any** thymocyte acquire receptors for self Ag will be killed by apoptosis (programmed cell death) this process called negative selection.
- In medulla positive selection occur when cells acquire molecules (receptors) by which recognized Ags in association with class I MHC and class II MHC molecules.
- These two processes negative and positive selection are called T-cells educations.

B- Bone marrow:

Bone marrow is the primary site for the production of all circulating blood cells in adults, including immature lymphocytes. It also serves as the main site for **B cell maturation**.



2. Secondary lymphoid organs:

- **Lymphocytes** become **functionally active** in the secondary lymphoid organs.
- These organs are small and underdeveloped at birth, but they gradually enlarge and mature as the individual ages.
- The secondary lymphoid organs include:

1. Spleen.
2. Lymph nodes.
3. Mucosal associated lymphoid tissues (MALT),

such as: gut associated lymphoid tissue (GALT).

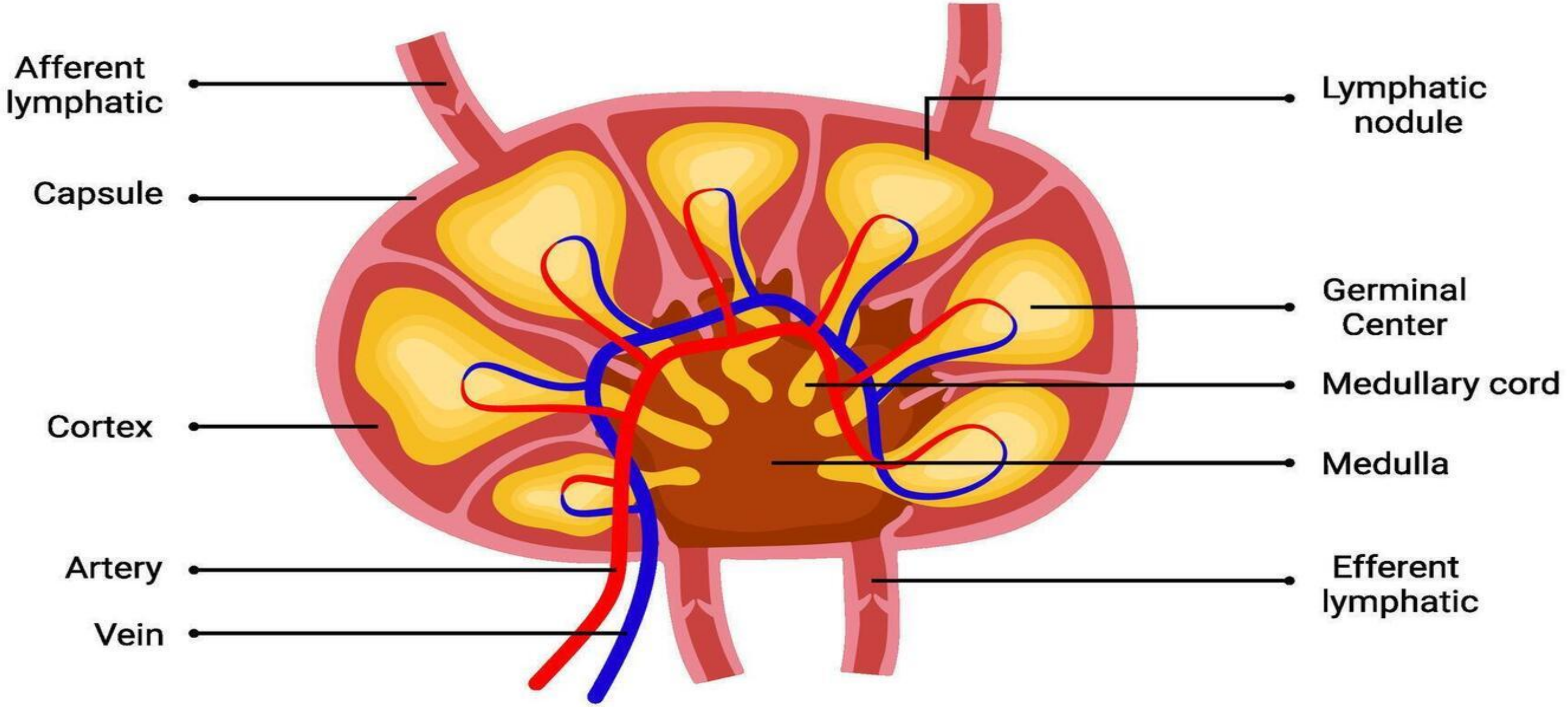
1- Spleen:

- Acts as a primary site for immune responses against antigens present in the bloodstream.
- Serves as a **reservoir** for immune cells, including lymphocytes and macrophages.
- **Produces** important immune molecules, such as components of the complement system.
- Functions in blood **filtration** by **removing** old or damaged red blood cells and eliminating pathogens from circulation.

2- Lymph Nodes:

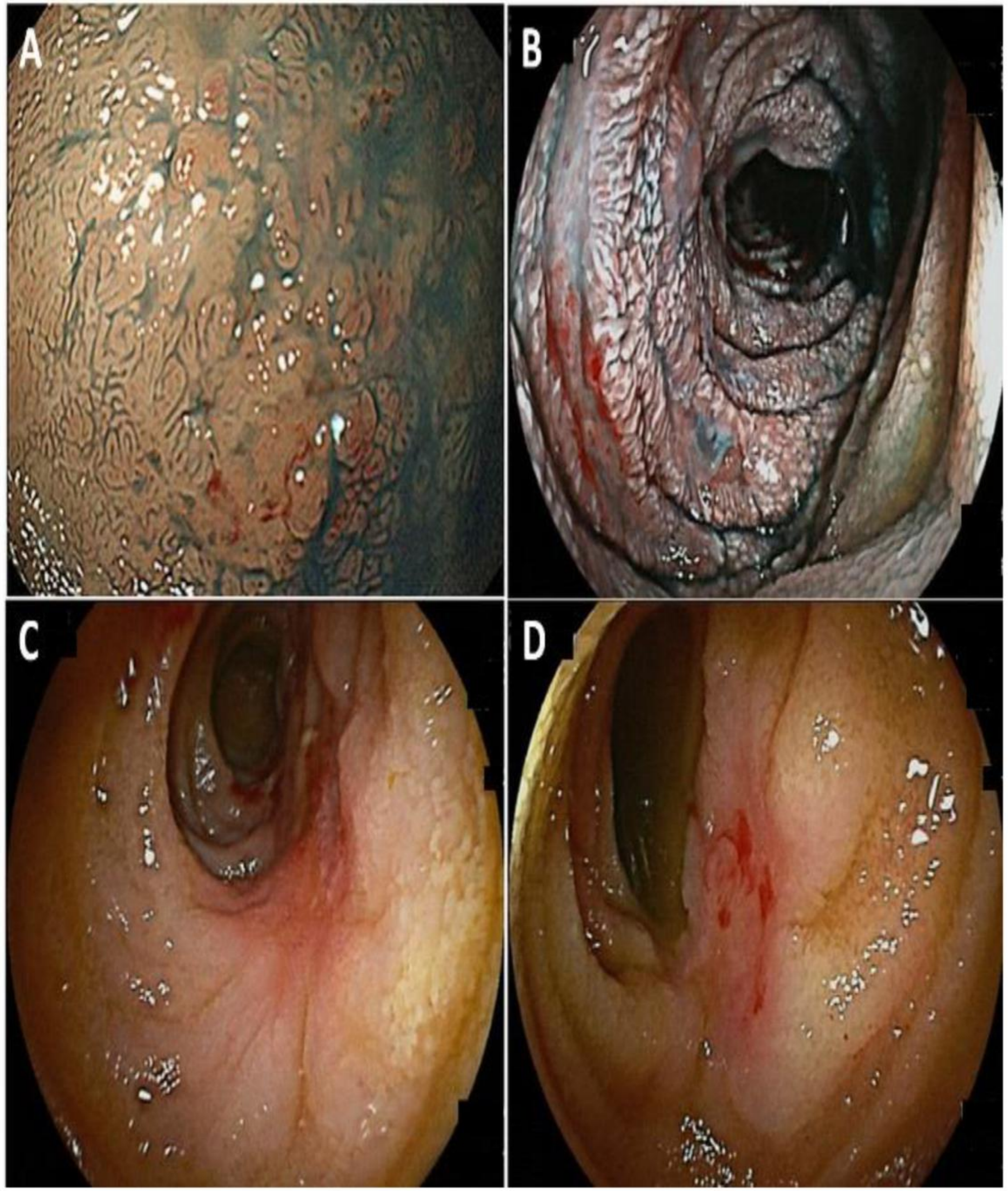
- Serve as key sites for initiating immune responses to antigens transported through the lymphatic system.
- Function to filter lymph, removing and destroying foreign antigens.
- Provide a habitat for immune cells, including **T and B lymphocytes**.
- Facilitate the recirculation of lymphocytes, enabling them to monitor the body for invading antigens.

Structure of Lymph Node



3. Mucosa-Associated Lymphoid Tissue (MALT):

The MALT in the gastrointestinal and respiratory tracts contains **lymphocytes** and **antigen-presenting cells** that trigger immune responses against antigens encountered through **ingestion** or **inhalation**.



Cells of the Immune System and their origin:

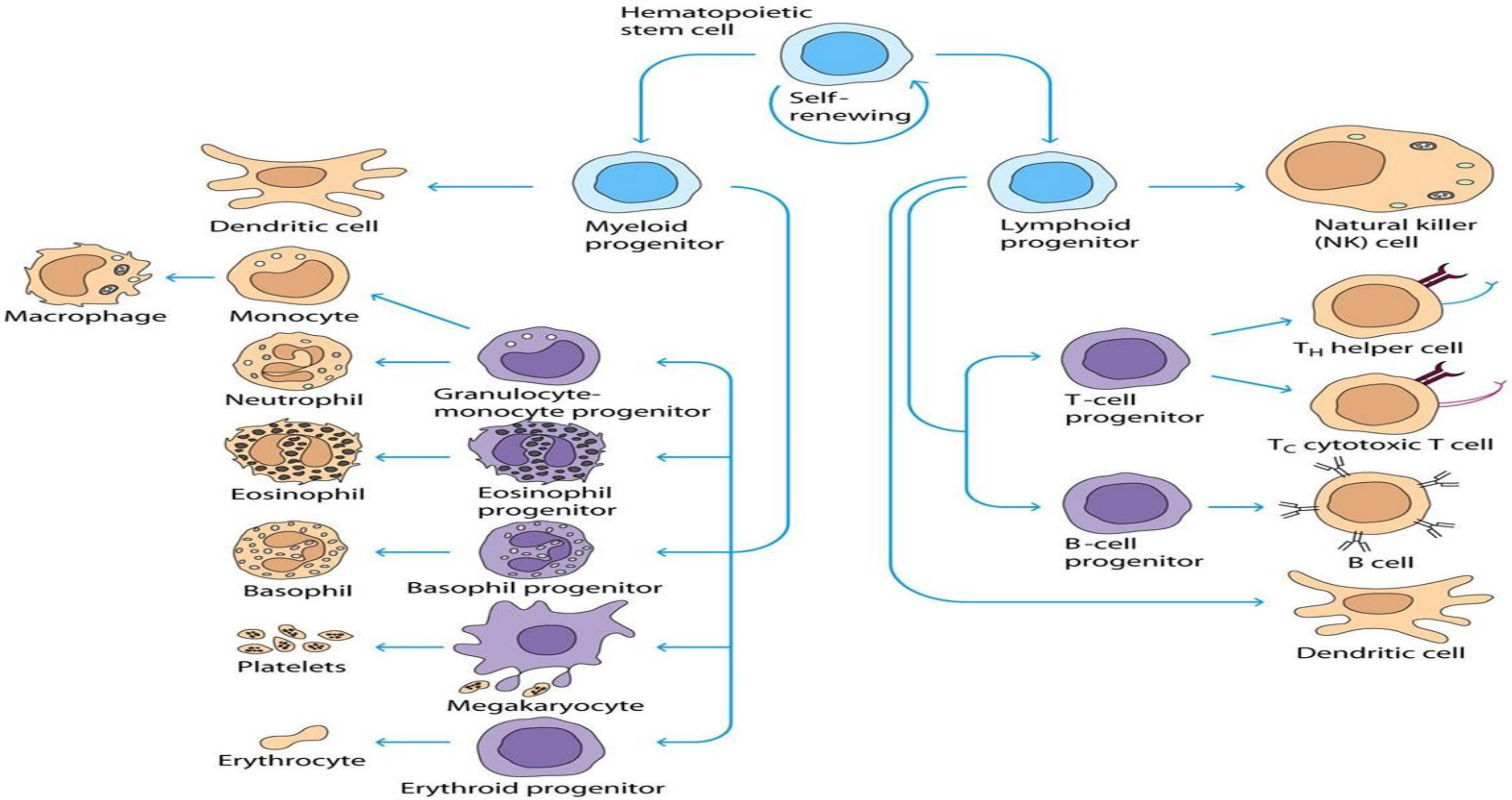
- Stem cells of the immune system originate from the **yolk sac** during the first six weeks of gestation. After this period, the **liver** takes over as the primary site of hematopoiesis. Eventually, the **bone marrow** becomes the main site for the production and proliferation of stem cells, under the influence of specific **hormones and enzymes**.
- These stem cells differentiate into two main lineages:

1- Lymphoid Series:

- T lymphocytes (T cells)
- B lymphocytes (B cells)
- Natural Killer (NK) cells

2- Myeloid Series:

- Red blood cells (RBCs)
- Monocytes
- Granulocytes (neutrophils, eosinophils, basophils)


Hematopoietic stem cell
Self-renewing
Myeloid progenitor
Lymphoid progenitor
Dendritic cell
Natural killer (NK) cell
Macrophage
Monocyte
Neutrophil
Granulocyte-monocyte progenitor
T_H helper cell
T-cell progenitor
T_C cytotoxic T cell
Eosinophil
Eosinophil progenitor
B-cell progenitor
B cell
Basophil
Basophil progenitor
Dendritic cell
Platelets
Megakaryocyte
Erythrocyte
Erythroid progenitor

A. Lymphocytes:

- Lymphocytes are mononucleate, no granular leukocytes of lymphoid tissue participating in immunity.
- They are found in blood, lymph and lymphoid tissue such as spleen, lymph nodes, tonsils and peyer's patches.
- They are spherical or oval in shape and arise from haemopoetic stem cells.
- They are classified into **two main types**:(T and B lymphocyte).

1-T-Lymphocytes:

- T lymphocytes constitute approximately **70%** of the total lymphocyte population.
- All T cells express **CD3** molecules and **T cell receptors (TCRs)** that recognize antigens presented by **MHC class I** or **MHC class II** molecules.
- **There are several distinct subtypes of T cells, including:**

1-T-Initiator/Inducer Cells: Express **$CD4^+$** on their surface.

2-T-Helper (Th) Cells: Also **$CD4^+$**; they stimulate B cells to produce antibodies and are further classified into **Th1** and **Th2** types.

3-T-Delayed Hypersensitivity (Tdh) Cells: Express **$CD4^+$** on their surface and are involved in delayed-type hypersensitivity reactions.

4-T-cytotoxic (Tc): Express $CD8^+$, destroy virus-infected and tumor cells.

5-T-memory: express **$CD4^+$** and/or **$CD8^+$** markers and play a key role in the **secondary immune response**.

2. B-lymphocytes:

B lymphocytes mature in the **bone marrow** and are responsible for **humoral immunity**, which involves the production of **antibodies** that target specific antigens.

B-Cell Development:

1-Pro-B Cell: Expresses **CD45** and **CD19** on its surface. This stage marks the **beginning of differentiation**, and the cell does not yet produce antibodies.

2-Pre-B Cell: Contains **intracytoplasmic μ (mu) chains**, indicating the **initiation of antibody synthesis**.

3-Immature B Cell: Displays **surface IgM** only.

4-Mature B Cell: Expresses both **surface IgM and IgD** molecules.

5-Activated B Cell: Upon activation, the B cell differentiates into a **plasma cell**, which secretes **antibodies** (immunoglobulins).

3. Natural killer cells (NK cells):

- They form the third population of lymphocytes.
- The NK cells have 2 or 3 large granules in the cytoplasm.
- They are also called large granular lymphocytes.
- They destroy the cancer cells and cells infected with virus, are activated by stimulation interferon and interleukin-2.

Mode of killing:-

1-Perforin-Granzyme Pathway:

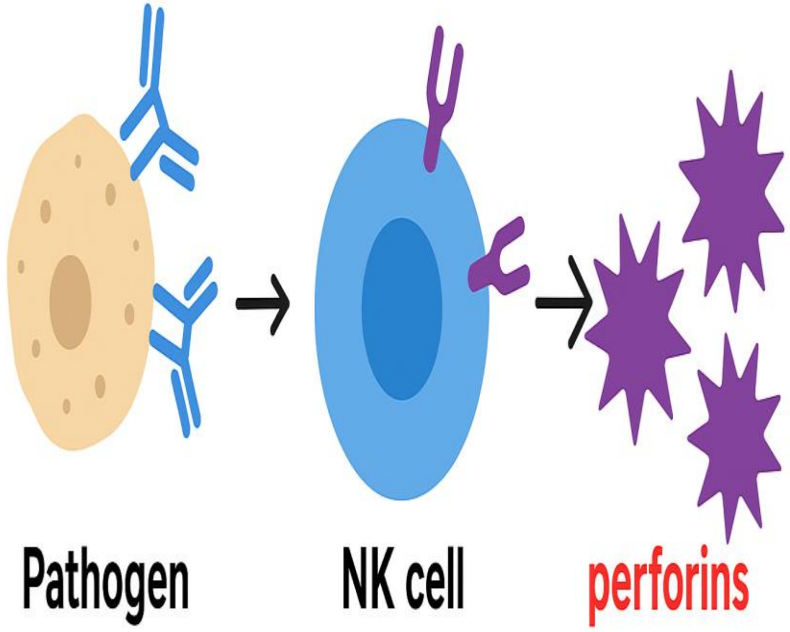
- NK cells release **perforins**, which create pores in the target cell membrane.
- **Granzymes** enter through these pores and trigger **apoptosis (programmed cell death)** in the target cell.

2-Antibody-Dependent Cellular Cytotoxicity (ADCC):

- In ADCC, natural killer (NK) cells recognize target cells that have been coated with antibodies (usually IgG).
- The Fc region of these antibodies binds to CD16 receptors on the surface of NK cells.
- This interaction activates the NK cell, leading it to release cytotoxic molecules such as perforin and granzymes, which induce apoptosis in the antibody-coated target cell.

Mode of killing:

Kill by Ab dependent cell-mediated cytotoxicity (ADCC).



B. Macrophages:

- Macrophages are large mononuclear phagocytic cells derived from monocytes.
- Macrophages are concentrated in lymph nodes, spleen, bone marrow and liver, their **function phagocytosis.**

C. Eosinophils:

- They are acidophilic leucocytes and are called eosinophils because eosin (acid dye) stains the granules of the cytoplasm of these cells .
- Their granules contain a high concentration of **hydrolytic enzymes**.

D. Basophils:

- These cells contain cytoplasmic granules that are stained by **basic dyes**.
- Their **basophilic granules** are rich in **heparin, histamine, serotonin**, and **platelet-activating factor (PAF)**.
- Play a key role in **allergic and inflammatory reactions**.

E. Neutrophils:

- Neutrophils form the major part of the white blood cell.
- They are motile, short lived cells with multi-lobed nucleus.
- Major function of the neutrophil is **phagocytosis**.

Complement system

- The **complement system** is a group of about **30 plasma proteins** that work together to **enhance (complement)** the immune response. These proteins circulate in the blood in an **inactive form** and become **activated by infection**.

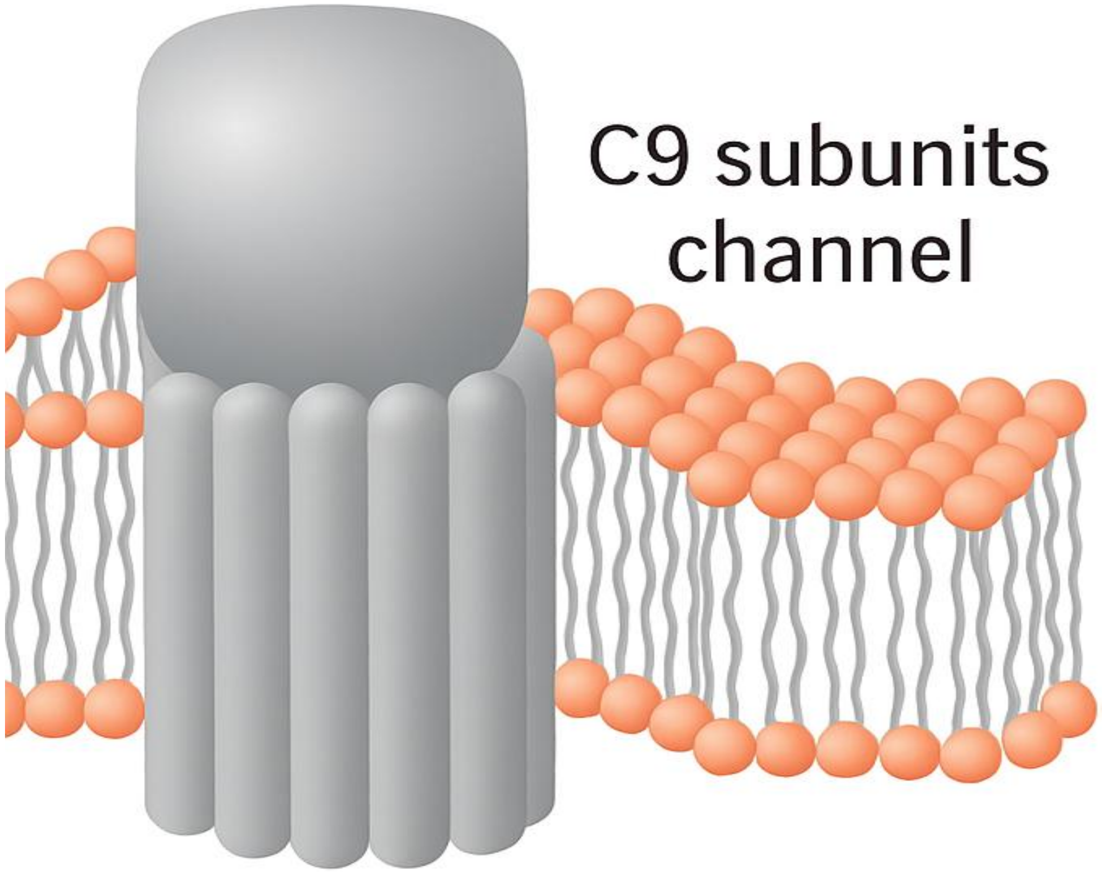
Functions:

1-Opsonization: coating microbes to make them easier for phagocytes to recognize and engulf (mainly by **C3b**).

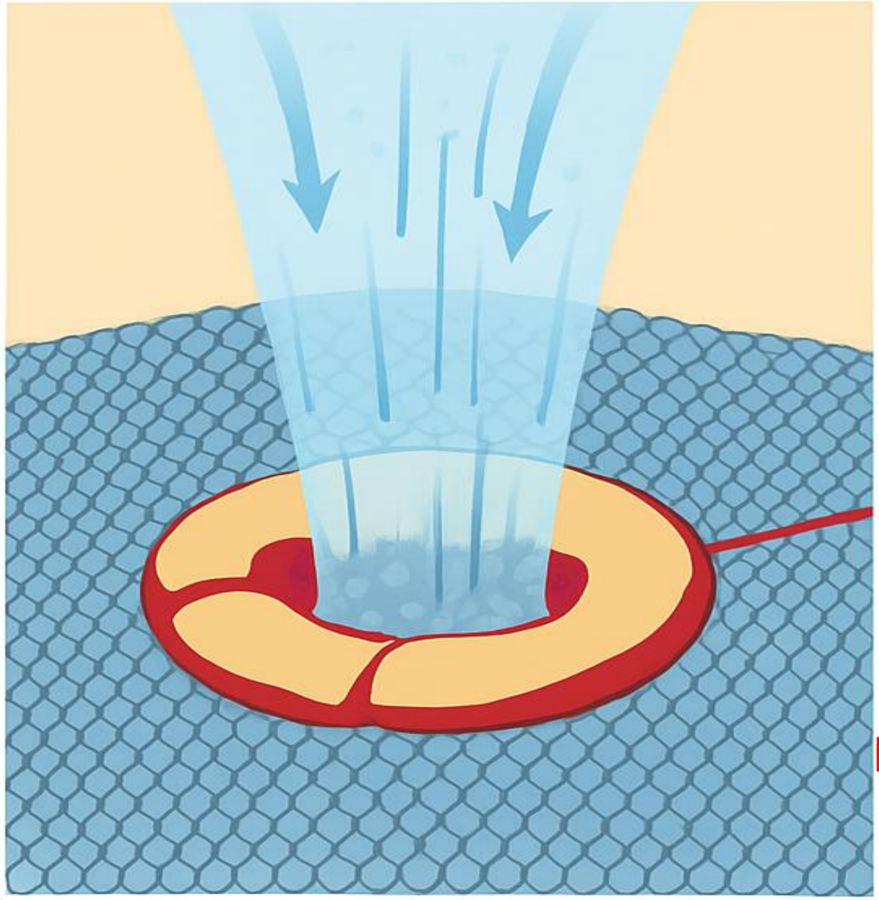
2-Chemotaxis and inflammation: attracting immune cells to the infection site (by **C3a and C5a**).

3-Cell lysis: forming the **Membrane Attack Complex (MAC)** that punctures the pathogen's membrane (C5b–C9).

4-Clearance of immune complexes: helping remove antigen–antibody complexes from circulation.



MAC Causes Cell Lysis and Death



Any Question?

Thank you